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Question Attempted	In this assignment, you will use the LendingClub dataset to: 1. Evaluate competing loan review policies using a simulated A/B testing framework, 2. Identify borrower segments using cluster analysis to understand heterogeneity in outcomes, and 3. Develop a prototype deep learning model to predict loan default and propose an improved loan review policy. This project is designed to build a fast prototype and initial analysis of the data. Further development and refinement would be completed later in a full project setting. Question 1: A/B Testing of Loan Review Policies (25%) Design and conduct a simulated A/B test to compare 2 loan approval policies. Your analysis should include: Q1a. Policy Definition Define the two loan review policies (Policy A and Policy B) being compared (e.g., using different approval thresholds based on income or existing debt amount, rule-based criteria, or loan grade cutoffs, etc.) and justify your choices. Q1b. Evaluation Metric and Hypotheses Specify one primary evaluation metric for evaluating the effectiveness of the loan review Policy A and Policy B (e.g., false positive rate, false negative rate, etc.) and explain your reasoning why it should be a primary evaluation metric. Include measurement details/formula. Page 1 of 5 Formulate the null and alternative hypotheses for comparing Policy A and Policy B. Q1c. Policy Comparison Using observed loan outcomes, compare the performance of the two policies. Include summary statistics, appropriate statistical comparisons, and interpretation of results in business terms. Q1d. Limitations Discuss the limitations of evaluating policies using observational data and how these affect causal interpretation. Question 2: Borrower Segmentation via Cluster Analysis (25%) Q2a. Feature Selection and Methodology Describe the variables used for clustering and justify their suitability. Justify the choice of clustering method and number of clusters. Q2b. Cluster Interpretation Profile and label each cluster using key characteristics and outcome statistics. Explain how clusters differ in borrower risk. Q2c. Policy Performance by Cluster Evaluate how Policy A and Policy B perform within each borrower cluster. Explain what additional insights clustering provides beyond aggregate results. Question 3: Deep Learning for Default Prediction (30%) Q3a. Prediction Task and Data Split Define the default prediction task and describe how the data is split into training, validation, and test sets. Q3b. Model Design and Training Describe the deep learning prototype model, including handling of variables, architecture, loss function, and class imbalance. Q3c. Model Evaluation Evaluate model performance using appropriate metrics and compare across the borrower clusters identified in Question 2. Question 4: Integrated Recommendation (10%) Q4a. Integration Explain how A/B testing, cluster analysis, and deep learning work together in this project. Q4b. Model Insights and Next Steps Provide some insights on loan review policy you have gained from the process. Briefly discuss the next steps of further developing these prototype models and how they should be evaluated in practice.
Have you used Artificial Intelligence (AI) in any part of this assignment?	Yes
If you have ticked “Yes” above, please briefly outline below which AI tool you have used, and what you have used it for. Please note, you must also reference the use of generative AI correctly within your assessment, in line with the guidance provided in your student handbook. This is to certify that the work submitted is entirely my own. All external sources and references have been properly documented. Artificial Intelligence tools were used in the development of this work as follows: Which AI: OpenAI ChatGPT and Anthropic Claude. Why: To support the coding process and to assist with reviewing the technical report during drafting. What each was used for: ChatGPT was used during the coding phase to generate example functions and cross-validate possible approaches, which were then reviewed, adapted, and independently applied to suit the context of this analysis. Claude was used to cross-check consistency between the written report and the underlying code, to identify logical gaps in the analytical narrative, and to suggest targeted edits for clarity and word count. All suggestions were reviewed, evaluated, and either accepted, modified, or rejected by myself.	

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Introduction

Consumer lending companies face a fundamental trade-off: approving too many loans increases default exposure, while approving too few sacrifices profitable volume. This report analyses LendingClub loan data from 2012–2013 using three sequential methods: A/B policy testing, borrower segmentation via clustering, and deep learning, to support an improved loan approval strategy.

A/B Testing of Loan Review Policies

Policy Definition and Justification

Two rule-based approval policies were defined (Table 1) using different thresholds for three key variables observable at origination: debt-to-income ratio (DTI), LendingClub grade (grade), and delinquencies in last two years (delinq_2yrs). These variables represent key dimensions of credit risk: repayment capacity, overall creditworthiness, and recent payment behaviour respectively, which are widely used in consumer credit risk assessment.

Table 1: Policy A vs Policy B - Approval Criteria

Variable	Policy A (Lenient / Growth Oriented)	Policy B (Strict / Conservatory)
dti	≤ 30	≤ 20
grade	{A–F}	{A–C}
delinq_2yrs	≤ 1	$= 0$

Evaluation Metric and Hypotheses

The primary Overall Evaluation Criterion (OEC) is average net return per application, defined as: $\text{net_return} = \text{total_pymnt} + \text{recoveries} - \text{collection_recovery_fee} - \text{funded_amnt}$

Rejected applications are assigned a return of zero. This metric was selected because it inherently balances both types of error: approving a defaulter who generates loss, while rejecting a creditworthy borrower generates zero return. The hypotheses are:

H_0 : $\text{net_return_A} = \text{net_return_B}$ (no difference between policies)

H_1 : $\text{net_return_A} \neq \text{net_return_B}$ (return differs between policies - two-sided Welch t-test).

Secondary metrics capture the components of that trade-off: FNR (opportunity cost of over-restriction), bad rate among approved loans (portfolio quality) and FPR (share of defaulters approved).

Policy Comparison Results

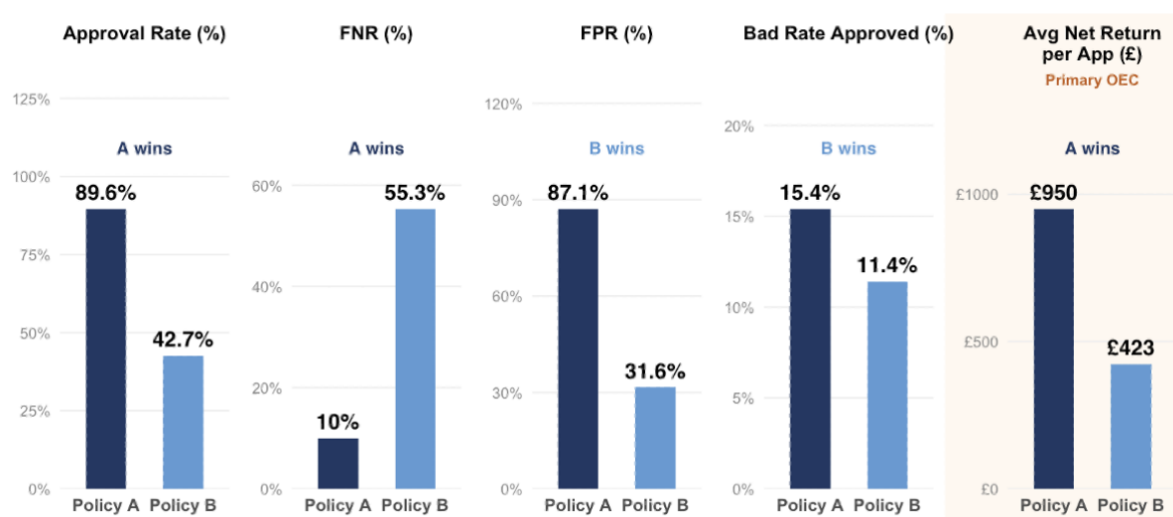
Policy A generates a higher average net return per application than Policy B (£949.65 vs £423.49), a difference of £526.16. A Welch t-test confirms this is highly significant ($p < 2.2 \times 10^{-16}$; 95% CI: [£471.77, £580.54]), establishing Policy A as the financially superior policy on the primary OEC. The effect size is negligible (Cohen's $d = 0.17$), reflecting high variance introduced by zero returns on rejected applications rather than a weak underlying difference, the £526 absolute gap remains economically meaningful.

The secondary metrics explain the magnitude of this gap (Table 2 & Fig 1). Policy B's approval rate of 42.7% versus 89.6% for Policy A means the majority of the £526 difference is driven by foregone revenue on rejected applications. Policy A's much lower FNR (10.0% vs 55.3%; $p < 2.2 \times 10^{-16}$) confirms it retains substantially more creditworthy borrowers. Policy B does produce a lower bad rate among approved loans (11.4% vs 15.4%) and lower FPR (31.6% vs 87.1%), confirming a safer portfolio, but this risk reduction is insufficient to offset the £526 per-application revenue penalty.

Table 2: Policy A vs Policy B — Summary Results

Metric	Policy A	Policy B
Approval Rate	89.6%	42.7%
FNR	10.0%	55.3%
Bad Rate (approved)	15.4%	11.4%
FPR	87.1%	31.6%
Avg Net Return / Application	£949.65	£423.49

Figure 1: Policy A vs Policy B — Metrics Comparison



Limitations

This analysis is based on historical data rather than a randomised experiment, so causal conclusions are limited, only issued loans are observed. Also, the profit measure excludes cost of capital, servicing costs, or discounting. Additionally, results are based on 2012–2013 data and may not generalise to different economic conditions.

Borrower Segmentation via Cluster Analysis

Feature Selection and Methodology

Eight financial variables were selected for clustering: loan amount, interest rate, annual income, DTI, number of open credit lines, revolving balance, revolving utilisation rate, and total accounts to capture three distinct dimensions of borrower risk: borrowing capacity, existing debt load, and credit behaviour. Critically, the target variable (loan_is_bad) and LendingClub grade were excluded to prevent leakage. Annual income was log-transformed

(log1p) to reduce the distorting influence of high-income outliers on Euclidean distance calculations.

Multivariate outliers were identified using Mahalanobis distance and removed at a chi-squared threshold of $p < 0.001$, eliminating approximately 1% of observations. All variables were then z-score standardised to ensure comparability across different measurement scales. To select k , hierarchical clustering was applied to a 2,000-observation sample using Ward's linkage method. Both the elbow plot and silhouette analysis confirmed $k = 4$ as the optimal number of clusters, balancing statistical fit with interpretability. K-means was then applied to the full dataset using hierarchical cluster centres as initialisations.

Cluster Interpretation

Table 3 summarises four borrower segments with clear differences in debt structure and default performance.

Prime Borrowers have relatively solid incomes and lowest DTI, revolving utilisation, and interest rates. Their default rate is lowest among all groups showing financially stable behaviour.

High Debt-Burden borrowers have income levels similar to Prime Borrowers but carry the highest DTI. Despite their relatively stable income, their default rate remains elevated, suggesting that heavy existing obligations constrain repayment capacity.

Wealthy but Aggressive borrowers earn the highest incomes but also take on the largest loans and maintain very high utilisation. The market appears to price this risk accordingly, as they face the highest interest rates and a relatively high default rate despite their strong income levels.

The Strugglers represent the largest and most financially pressured segment. They have the lowest income and very high revolving utilisation. Although their DTI is only moderate, they show the highest default rate, indicating liquidity stress concentrated in revolving credit.

Table 3: Borrower Segment Profiles

Segment	N	Avg Income (£)	Avg Loan (£)	Avg DTI (%)	Int Rate (%)	Util (%)	Default (%)
Prime Borrowers	11,046	69,898	11,207	10.8	10.5	33.1	9.8
High Debt-Burden	10,934	69,685	12,821	21.9	13.4	55.3	15.3
Wealthy but Aggressive	10,272	102,912	24,111	17.9	16.8	70.9	17.8
The Strugglers	17,142	48,742	10,062	18.5	15.0	70.7	18.3

Policy Performance by Cluster

Table 4 and Fig 2 compare Policy A and Policy B within each segment where policy performance varies:

- For Prime Borrowers, Policy A approves more loans with little increase in default risk, so Policy B rejects many profitable customers.
- For High Debt-Burden borrowers, Policy B reduces approvals but also lowers risk, showing that DTI rules work for this segment.
- For Wealthy but Aggressive borrowers, Policy B's grade-based restriction substantially reduces approvals (29.8% vs 88.6%) and FPR (17.0% vs 85.6%), but the segment still carries a 17.8% default rate, suggesting that income and grade alone do not fully capture the leverage and utilisation risk in this group.
- For The Strugglers, Policy B rejects many borrowers, but default rates remain high, indicating that utilisation risk is not captured by either policy.

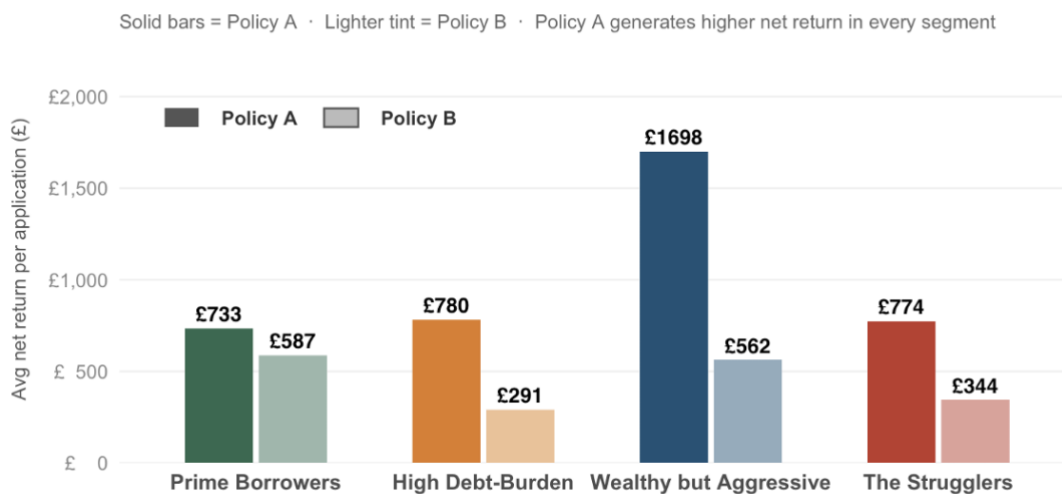
Figure 2: Average Net Return per Application by Borrower Segment

Table 4: Policy Performance by Borrower Segment

Segment	A: Approval (%)	A: FPR (%)	A: FNR (%)	B: Approval (%)	B: FPR (%)	B: FNR (%)
Prime Borrowers	96.2	94.7	3.6	77.7	71.5	21.6
High Debt-Burden	81.3	78.3	18.1	27.3	19.7	71.3
Wealthy but Aggressive	88.6	85.6	10.8	29.8	17.0	67.4
The Strugglers	90.9	89.9	8.9	38.2	31.7	60.3

Figure 2 confirms that Policy A generates higher net return per application in every segment most substantially for Wealthy but Aggressive where Policy B's strict grade restrictions eliminate high-value loans, and for The Strugglers where Policy B's conservatism costs significant revenue without meaningfully reducing defaults. Even in High Debt-Burden, where Policy B's DTI rules demonstrably reduce risk, the financial penalty remains severe. This confirms that a single rule-based policy cannot perform optimally for all borrower types and motivates the use of predictive models.

Deep Learning for Default Prediction

Prediction Task, Feature Selection and Data Split

The prediction task is binary classification: given borrower and loan characteristics available at origination, predict whether a loan will default (`loan_is_bad` = 1) or be fully repaid (`loan_is_bad` = 0).

Seventeen origination-time features were selected, avoiding duplication and target leakage. The final feature set covers: borrower financials (`loan_amnt`, `annual_inc`, `dti`, `revol_bal`, `revol_util`), credit behaviour (`delinq_2yrs`, `inq_last_6mths`, `open_acc`, `pub_rec`, `total_acc`, `emp_length`), loan characteristics (`int_rate`, `term`, `grade`), and categorical context (`home_ownership`, `purpose`, `verification_status`). After dropping rows with missing values, 48,168 complete cases remain, expanding to 32 features after one-hot encoding.

The data were split into training (70%), validation (15%), and test (15%) sets using stratified random sampling. The validation set guides early stopping and hyperparameter selection; the test set is held out entirely until final evaluation, preventing hyperparameter choices from contaminating reported performance.

Model Design, Architecture and Training

Variable encoding was applied before model training. Ordinal variables with a natural ordering were integer-encoded: `grade` (A=1 to G=7) and `term` (36 months = 0, 60 months = 1). Nominal categorical variables with no natural ordering, `home_ownership`, `purpose`, and `verification_status` were one-hot encoded. All numeric features were z-score standardised on the training set only.

The class imbalance (~85% good loans, ~15% defaults) was addressed through random undersampling of the majority class(class 0) producing a balanced 50/50 dataset.

Undersampling was preferred over synthetic oversampling because the dataset is sufficiently large to absorb the majority-class reduction without information loss, and avoids introducing synthetic minority observations that may not reflect real borrower profiles.

The model is a Dense Neural Network (DNN) with a funnel architecture: three hidden layers of 128, 64, and 32 neurons respectively, each followed by BatchNorm1d, ReLU activation, and Dropout for regularisation and training stability. Batch normalisation stabilises gradient flow by normalising activations within each mini-batch, addressing internal covariate shift. ReLU provides non-linearity without the vanishing gradient problem of sigmoid or tanh. Dropout randomly zeroes a fraction of neurons per pass, preventing co-adaptation and acting as an ensemble regulariser. The output layer returns a single raw logit; BCEWithLogitsLoss using pos_weight = 1.0 applies the sigmoid internally for numerical stability. Training used AdamW with ReduceLROnPlateau decay and early stopping by validation loss; five hyperparameter tuning rounds were conducted (see Table 5).

Table 5: Hyperparameter Tuning Results

Parameter	Values Tested	Best Value	Best AUC
Architecture	64-32 / 128-64-32 / 256-128-64 / 256-128-64-32	128-64-32	0.6709
Learning rate	0.01 / 0.005 / 0.001 / 0.0005 / 0.0001	0.005	0.6715
Weight decay	0 / 1e-5 / 1e-4 / 1e-3 / 1e-2	0.01	0.6710
Batch size	64 / 128 / 256 / 512	256	0.6709
Dropout rate	0.1 / 0.2 / 0.3 / 0.4 / 0.5	0.2	0.6703

Model Evaluation and Cluster Interpretation

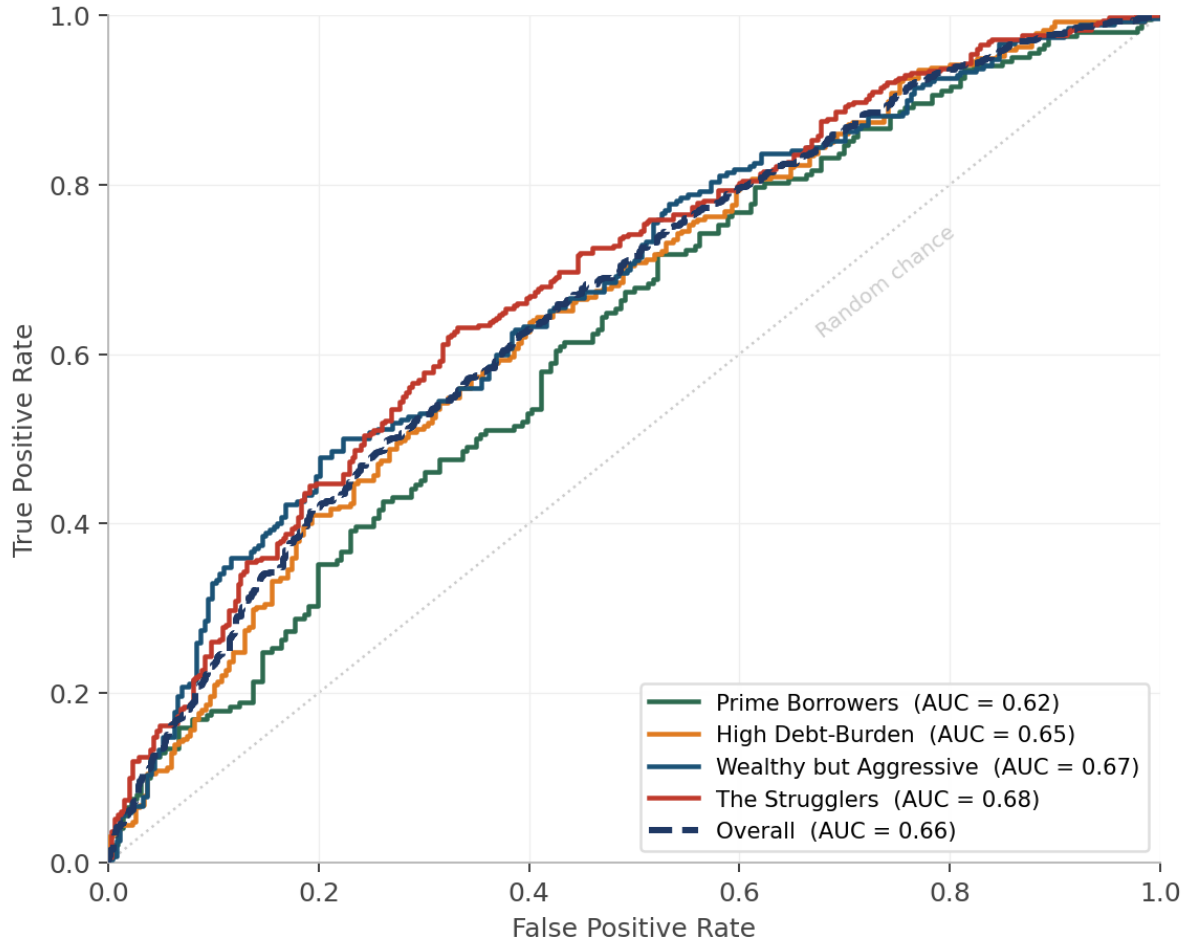
Table 6 and Figure 3 report model performance across the earlier borrower segments on the held-out test set. The primary metric is AUC-ROC, supplemented by recall, precision, and F1. High recall is commercially prioritised because undetected defaults generate direct credit losses.

The model achieved an overall AUC of 0.66, recall of 0.70, precision of 0.59, and F1 of 0.64. This exceeds random chance and demonstrates genuine discriminative power over origination-time features alone. With the dataset balanced 50/50 and the threshold of 0.5, the model correctly flags 70% of defaults while achieving 59% precision.

The segment-level results provide clear business insight. The Strugglers and Wealthy but Aggressive segments achieve the highest AUC (0.68 and 0.67), confirming the DNN captures the utilisation-driven and income-leverage risk patterns that neither policy rule encodes. High Debt-Burden borrowers show an AUC of 0.65, where DTI and grade provide a reliable signal. Prime Borrowers show the lowest AUC (0.62), reflecting the genuine difficulty of discriminating within an already low-risk population despite their post-undersampling default rate of 47%. These results confirm that a model-based approach adds value over rule-based policies, particularly for segments whose risk is multi-dimensional.

Table 6: Model Performance by Borrower Segment

Segment	N	Default %	AUC	F1	Precision	Recall
Prime Borrowers	428	47.20%	0.62	0.61	0.54	0.69
High Debt-Burden	565	52.20%	0.65	0.64	0.62	0.67
Wealthy but Aggressive	544	49.60%	0.67	0.63	0.58	0.70
The Strugglers	703	50.20%	0.68	0.66	0.60	0.73
Overall (test set)	2240	50%	0.66	0.64	0.59	0.70

Figure 3: ROC curves by borrower segment — DNN model

Integrated Recommendation

Integration

The three analytical methods form a sequential pipeline, each addressing limitations of the previous stage. The A/B test established that Policy A is the financially superior policy. However, this aggregate result conceals important heterogeneity: cluster analysis revealed that Policy A's advantage is not uniform across borrower types. For High Debt-Burden borrowers specifically, Policy B's stricter DTI rules substantially reduce FPR, suggesting blanket leniency carries real risk in that segment. However, even in that segment, Policy A still generates nearly three times the net return per application, confirming that the risk reduction

does not justify the financial cost. Crucially, neither policy captured revolving utilisation risk, the primary driver of default among The Strugglers, the largest and highest-defaulting segment. The deep learning model directly addresses this gap, achieving recall of 0.73 on that segment. Unlike rule-based policies which check variables independently, DNN learns non-linear interactions between revolving utilisation, income, and DTI simultaneously, allowing it to identify the liquidity stress pattern that neither policy threshold captures. Together, the three methods move from population-level rules to segment-level insights to individual-level probabilistic scoring, each layer adding explanatory power that the previous could not provide alone.

Insights and Next Steps

The combined analysis yields three clear policy insights. First, approval volume not default avoidance is the dominant driver of financial return, as Policy B's conservative approach sacrifices far more in foregone revenue (£526 per application) than it recovers through lower bad rates. Second, borrower risk is multi-dimensional: DTI-based rules reduce default risk for High Debt-Burden borrowers but at too high a revenue cost to justify, and are entirely blind to liquidity stress in The Strugglers, where revolving utilisation is the key risk signal. Third, the deep learning model (AUC=0.66) demonstrates genuine discriminative power beyond origination-time rule sets, particularly for segments whose risk involves interactions between income, leverage, and utilisation.

The recommended next step is to replace binary rule-based approval with a model-based threshold policy: approve applicants whose predicted default probability falls below a calibrated business threshold, with that threshold set by optimising expected net return rather than a fixed 0.5 cutoff. This would allow the company to retain high approval volume of Policy A while selectively screening out the highest-risk applicants that simple rules miss. Before full deployment, this model-based policy should be tested in a true randomised champion/challenger trial to obtain causal evidence of improvement and address the selection bias inherent in observational evaluation. Finally, the model should be monitored continuously post-deployment, with periodic retraining as borrower behaviour and macroeconomic conditions evolve beyond the 2012–2013 training window.